

Literature Status Note for arXiv:1705.05145

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Status

This is a lightweight literature-status packet, not a new proof. It records that two questions in Section 4 of Garcia-Lirola-Prochazka-Rueda Zoca, *A characterisation of the Daugavet property in spaces of Lipschitz functions*, arXiv:1705.05145, were answered in later papers.

Source Questions

The source paper asks in Section 4:

if M has property (Z) , must M be a length space?

The same section ends with the vector-valued problem:

$$\text{Lip}(M) \text{ Daugavet} \implies \text{Lip}(M, X) \text{ or } \mathcal{F}(M) \widehat{\otimes}_{\pi} X \text{ Daugavet?}$$

for a pointed metric space M and a Banach space X .

Supporting Answers

Aviles-Martinez-Cervantes, *Complete metric spaces with property (Z) are length spaces*, arXiv:1808.00715, explicitly identifies the property- (Z) problem as Question 1 from the source paper. Its Main Theorem states:

$$M \text{ complete metric} \implies M \text{ is length if and only if } M \text{ has property } (Z).$$

Thus the complete metric-space form used in the source's Lipschitz-free and Daugavet equivalences is already answered affirmatively.

Medina, *A characterisation of the Daugavet property in spaces of vector-valued Lipschitz functions*, arXiv:2305.05956, explicitly says that the vector-valued problem was asked as Question 1 in the source paper and gives a complete positive solution. Theorem 1 below records the substance of Medina's Theorem `theo:mainDauga`, and Corollary 1 records Medina's Corollary `cor:Daugavetvectorval`.

Theorem 1 (Medina, arXiv:2305.05956, Theorem `theo:mainDauga`). *If M is a complete length metric space and X is a Banach space, then $\text{Lip}(M, X)$ has the Daugavet property.*

Corollary 1 (Medina, arXiv:2305.05956, Corollary `cor:Daugavetvectorval`). *If M is a length metric space and X is a nonzero Banach space, then $(\mathcal{F}(M) \widehat{\otimes}_\pi X)^*$ has the Daugavet property; in particular, $\mathcal{F}(M) \widehat{\otimes}_\pi X$ has the Daugavet property.*

Combining this with the source theorem that, for complete M , $\text{Lip}(M)$ has the Daugavet property if and only if M is length, answers the vector-valued question for the complete-space setting of the source characterization.

Scope

The property-(Z) answer is recorded with the completeness hypothesis: this is the form explicitly posed and answered by arXiv:1808.00715 and the form needed for the source paper’s Banach-space equivalences. This packet does not assert a blanket resolution of every non-complete formulation of “length”.

The vector-valued answer is the special Lipschitz-free/vector-valued Lipschitz result from arXiv:1705.05145. It does not settle the broader Werner problem about arbitrary projective tensor products of Daugavet spaces.

References

- [1] L. Garcia-Lirola, A. Prochazka, and A. Rueda Zoca, *A characterisation of the Daugavet property in spaces of Lipschitz functions*, J. Math. Anal. Appl. 464 (2018), 473–492; arXiv:1705.05145.
- [2] A. Aviles and G. Martinez-Cervantes, *Complete metric spaces with property (Z) are length spaces*, arXiv:1808.00715.
- [3] R. Medina, *A characterisation of the Daugavet property in spaces of vector-valued Lipschitz functions*, arXiv:2305.05956.